

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

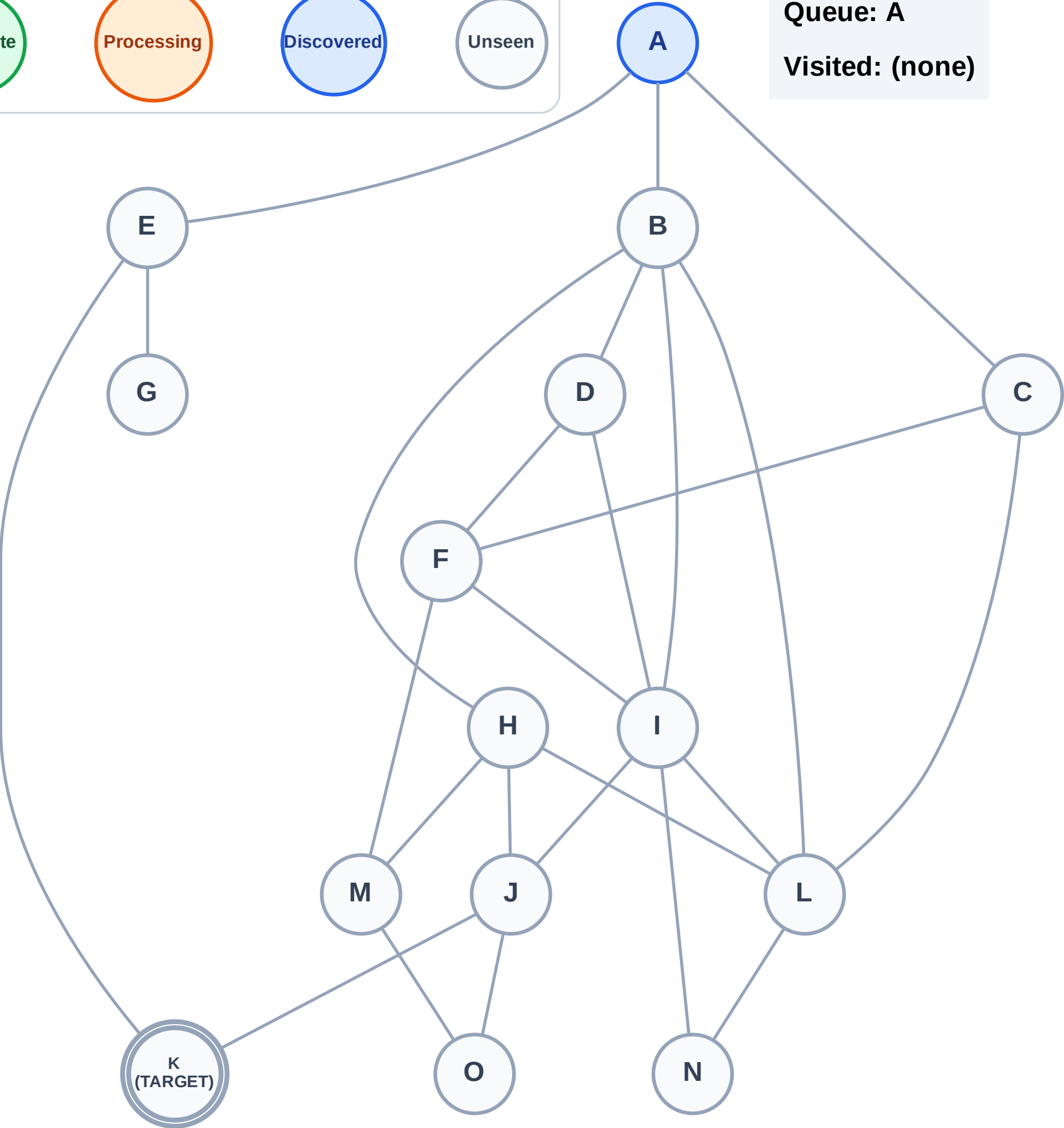
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

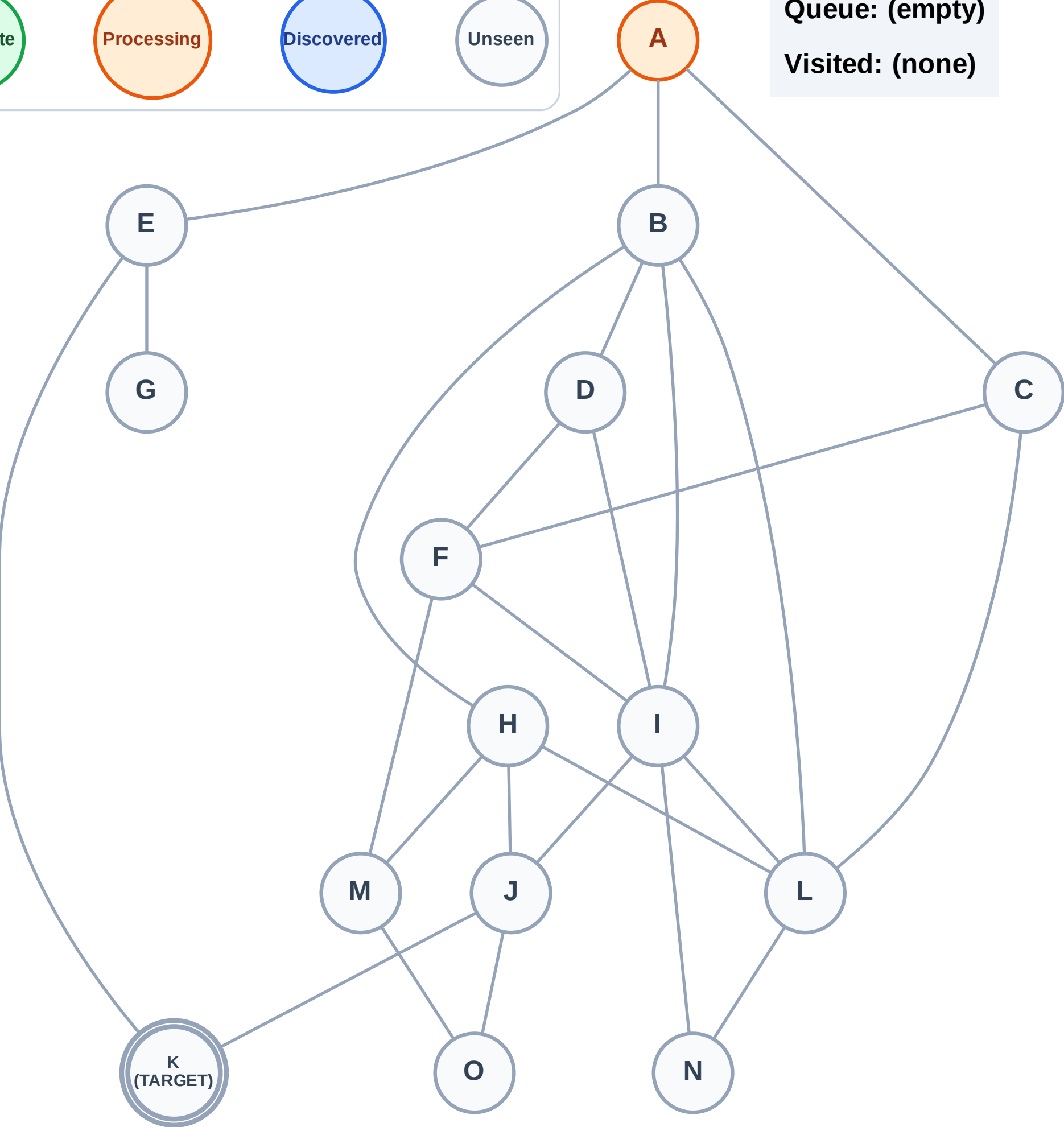
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, C, E from A

Legend

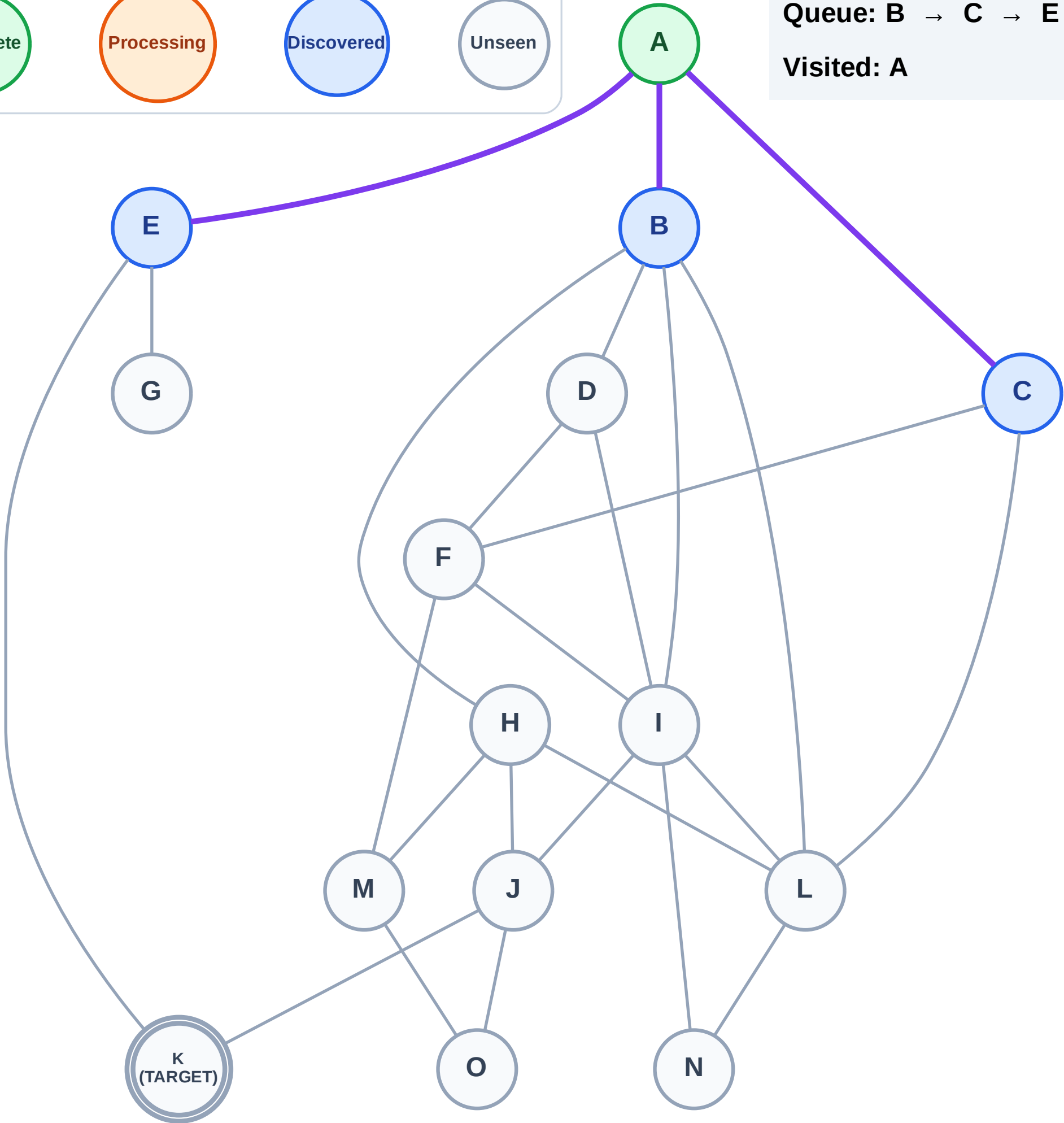
Complete

Processing

Discovered

Unseen

Queue: B → C → E
Visited: A



BFS Traversal

Step 4: Process node B

Legend

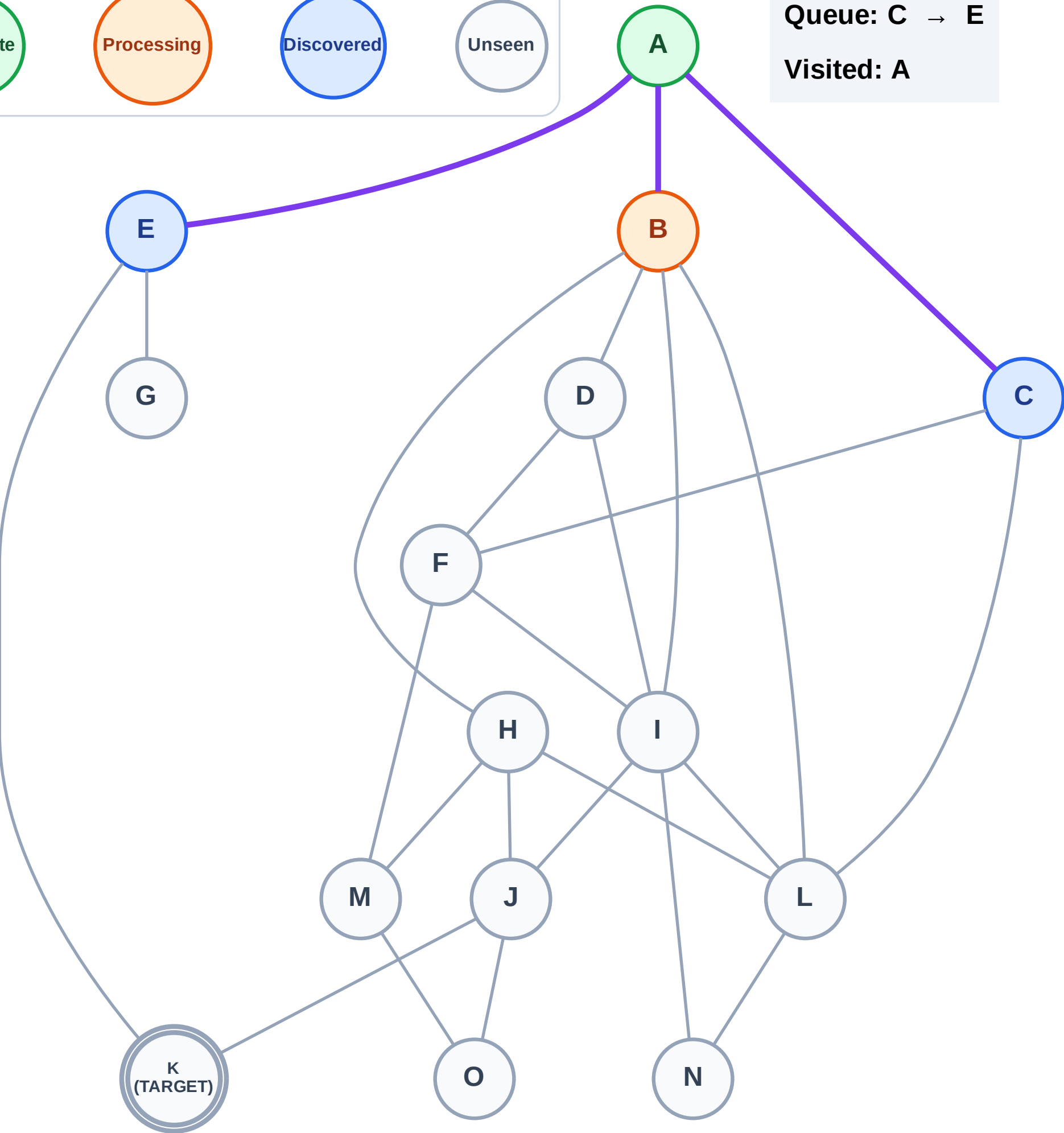
Complete

Processing

Discovered

Unseen

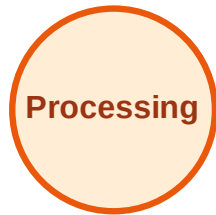
Queue: C → E
Visited: A



BFS Traversal

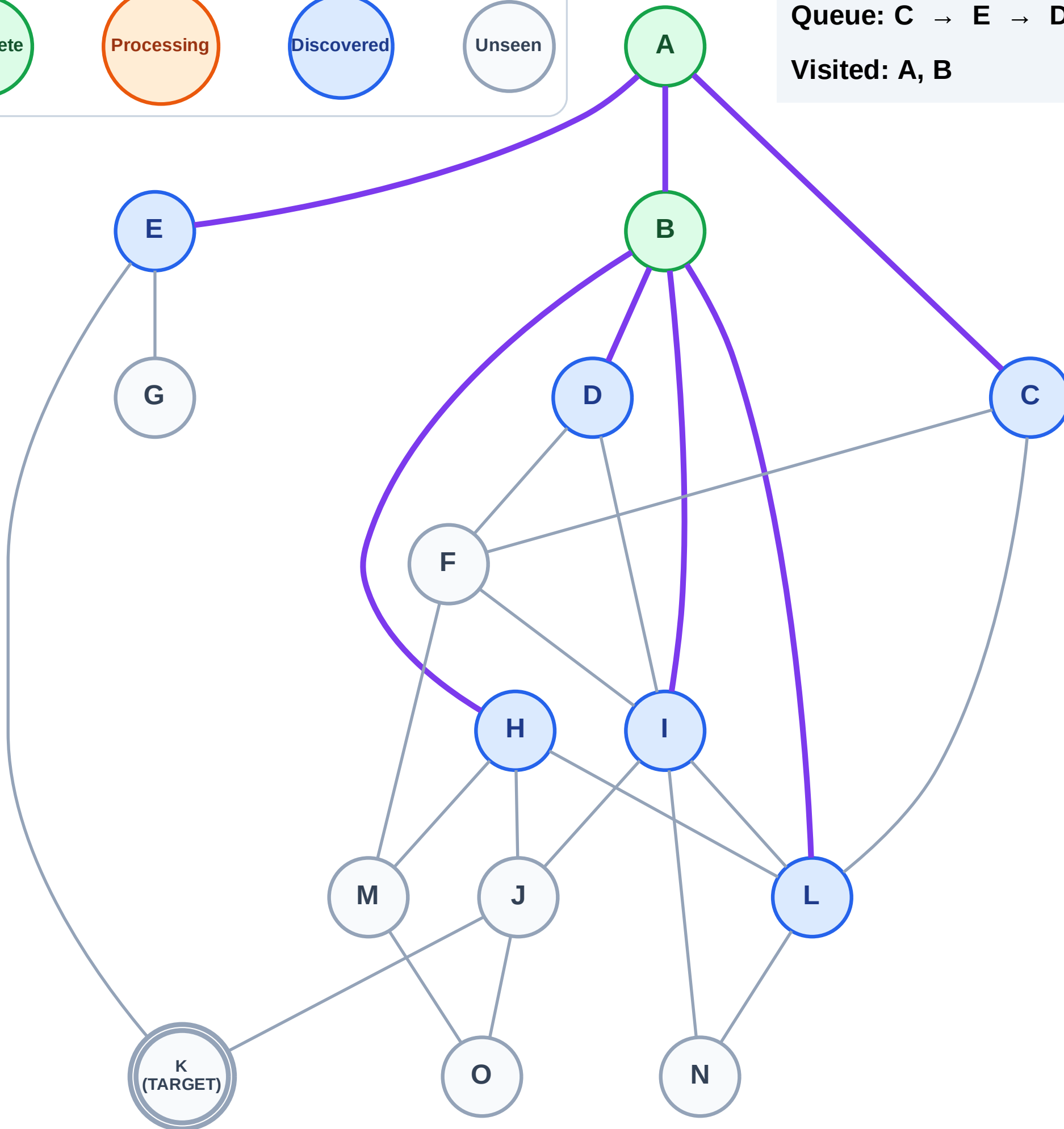
Step 5: Discover D, H, I, L from B

Legend

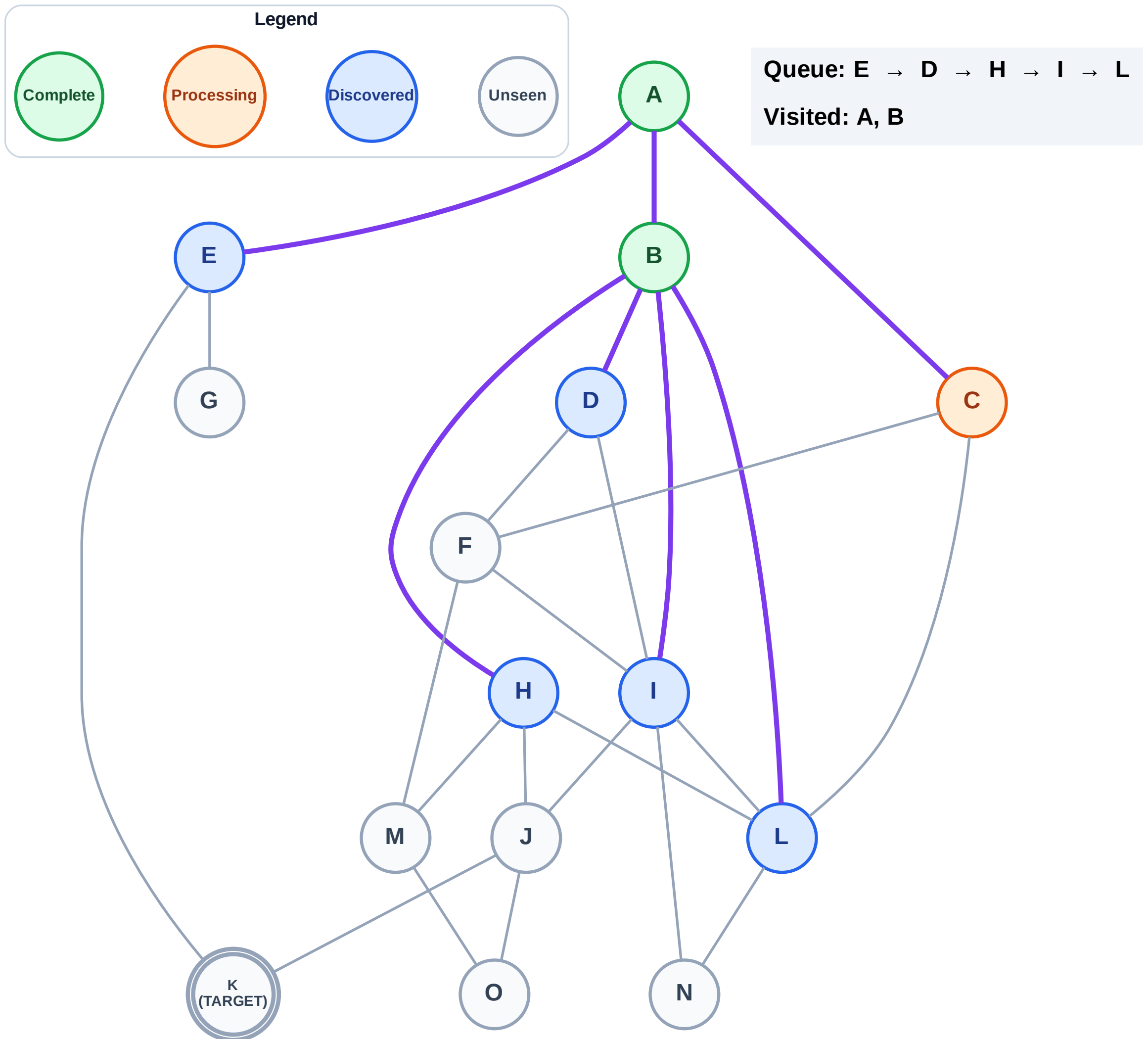


Queue: C → E → D → H → I → L

Visited: A, B



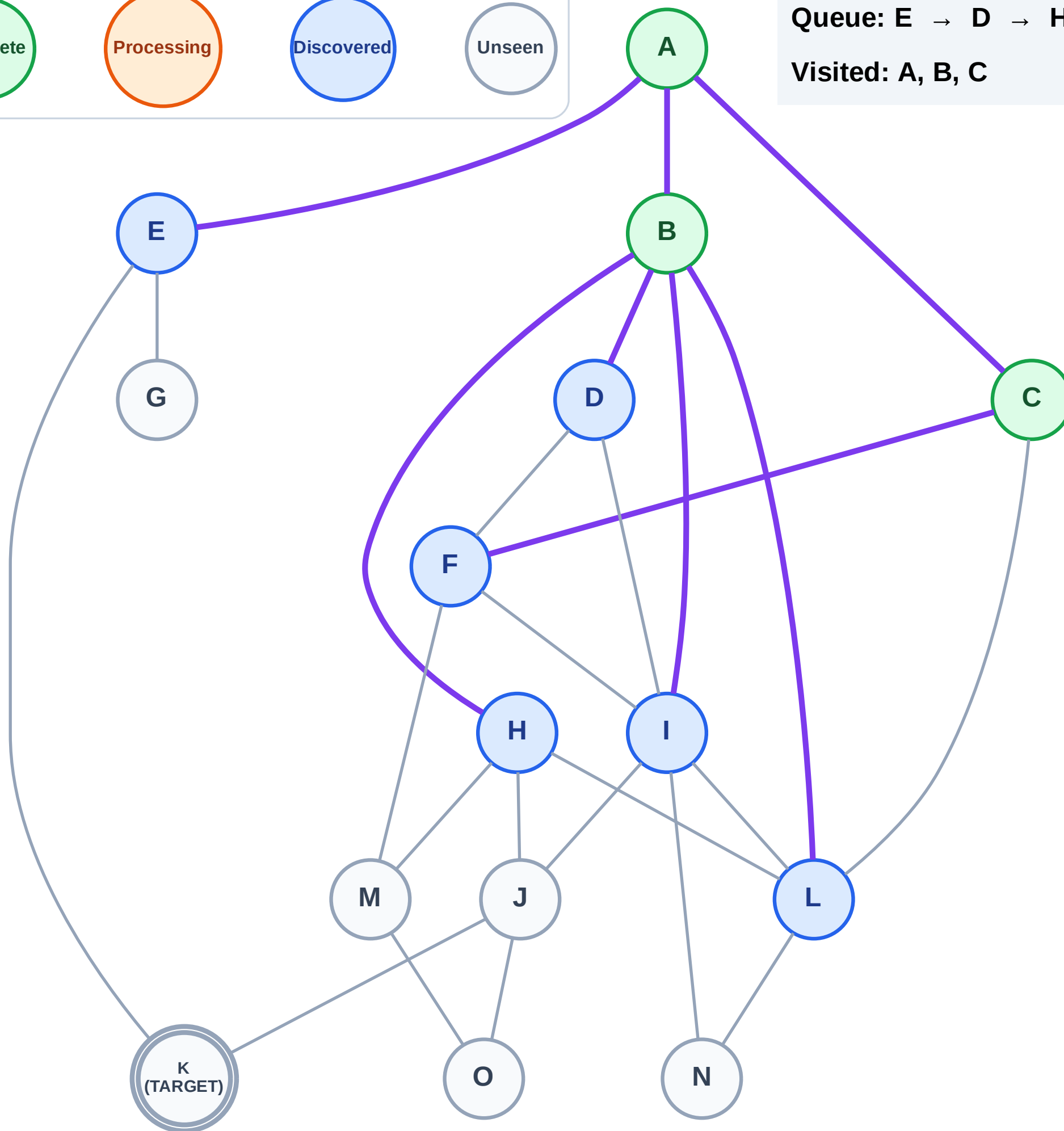
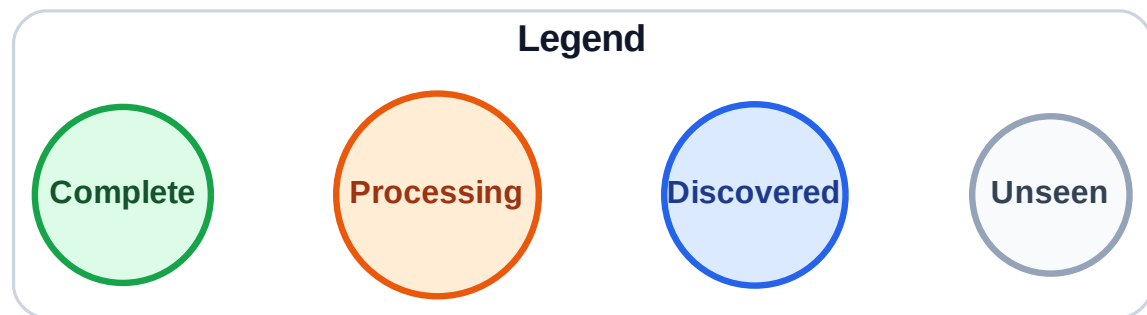
Step 6: Process node C



BFS Traversal

Step 7: Discover F from C

Queue: E → D → H → I → L → F
Visited: A, B, C



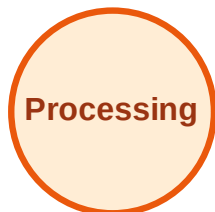
BFS Traversal

Step 8: Process node E

Legend



Complete



Processing



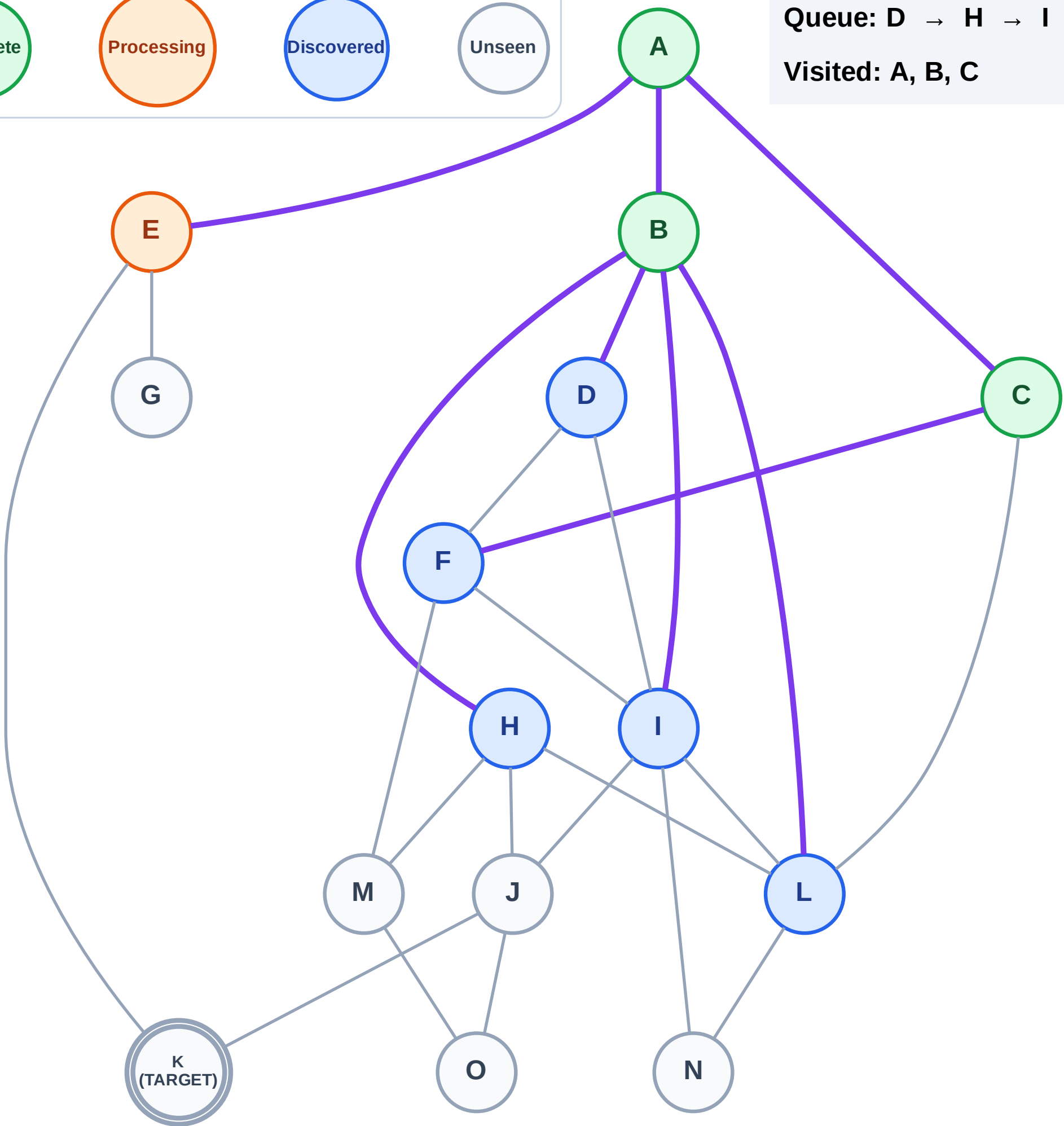
Discovered



Unseen

Queue: D → H → I → L → F

Visited: A, B, C



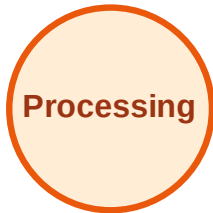
BFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → E → K

Legend



Complete



Processing



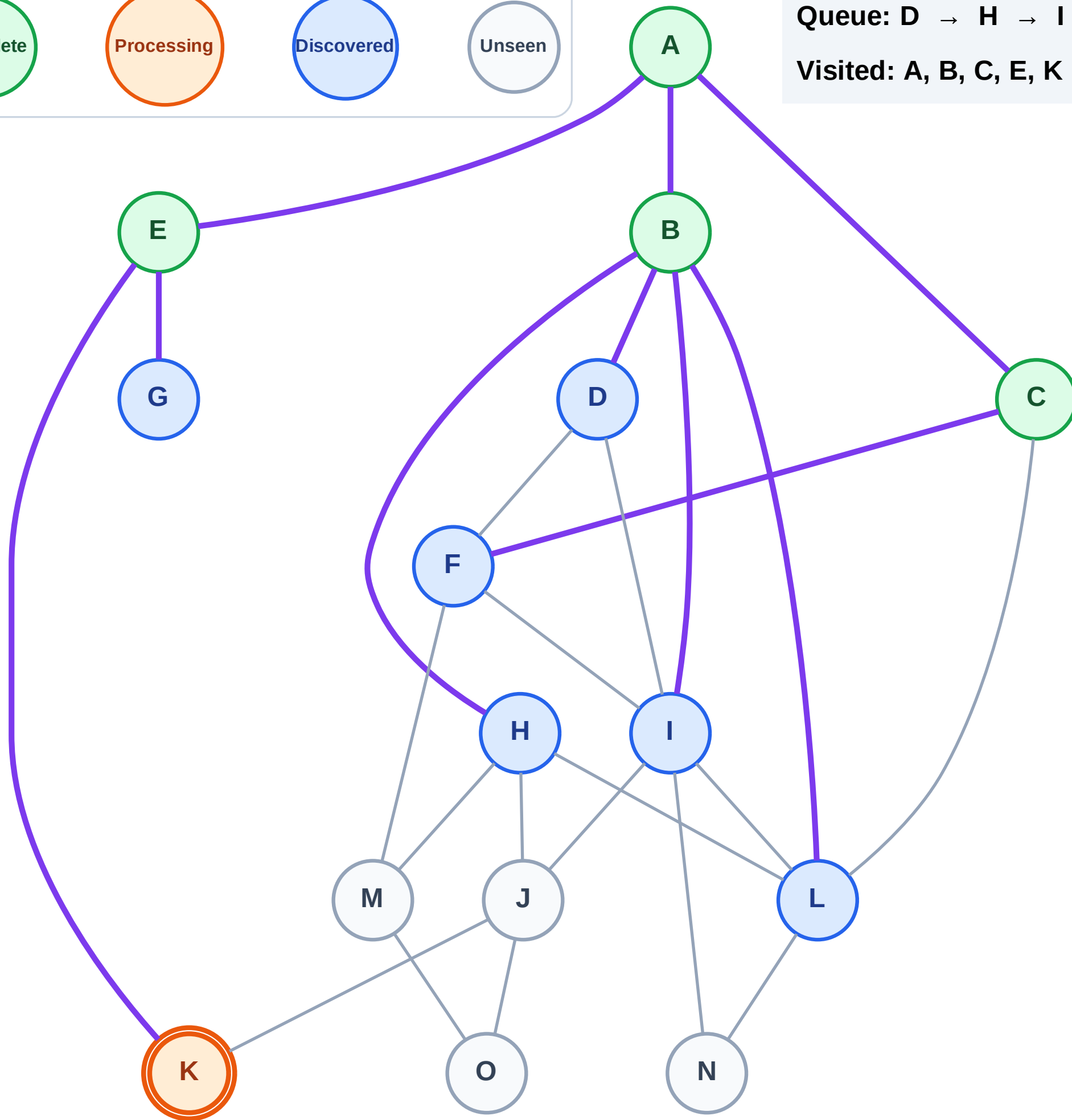
Discovered



Unseen

Queue: D → H → I → L → F → G

Visited: A, B, C, E, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

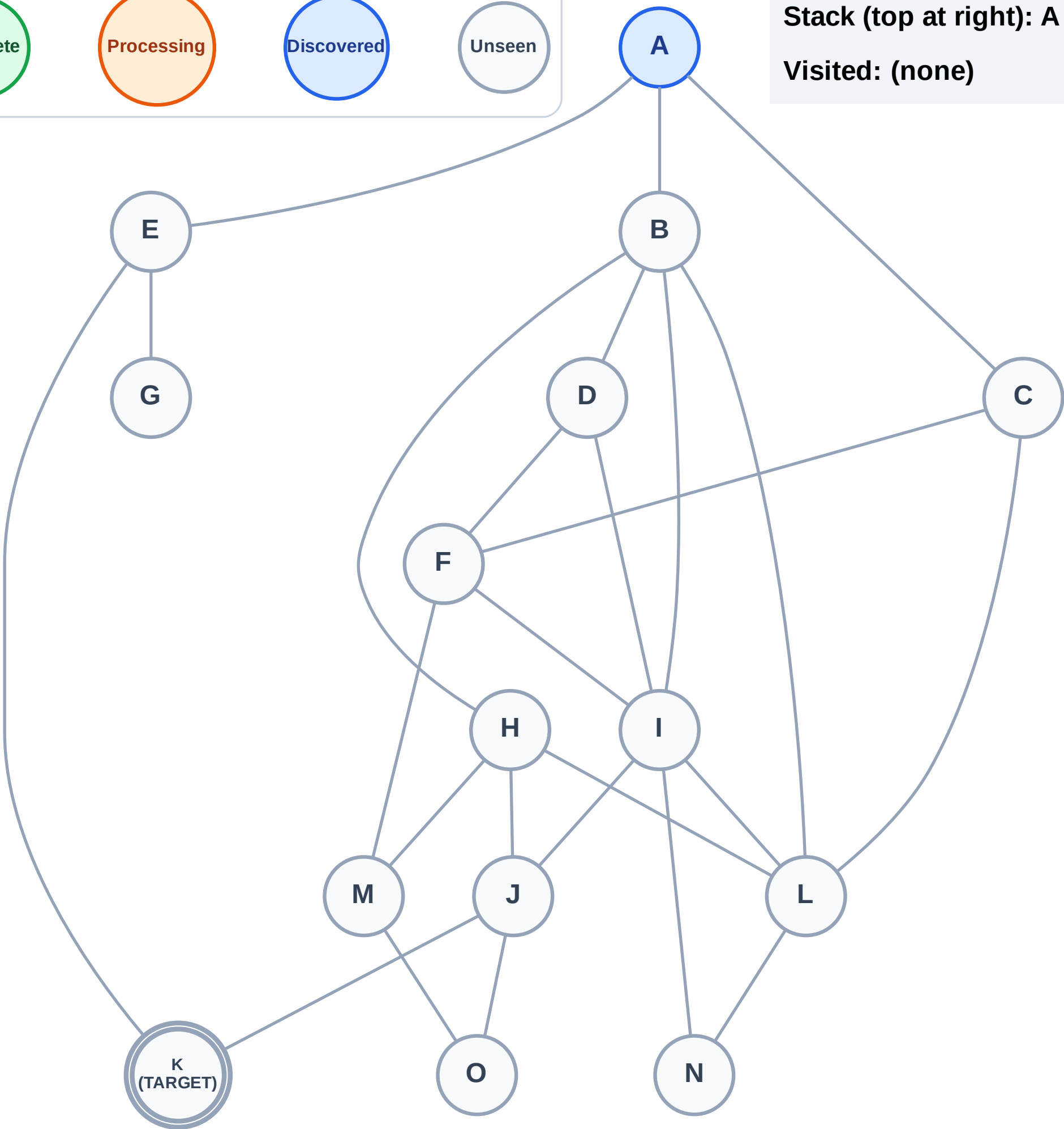
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

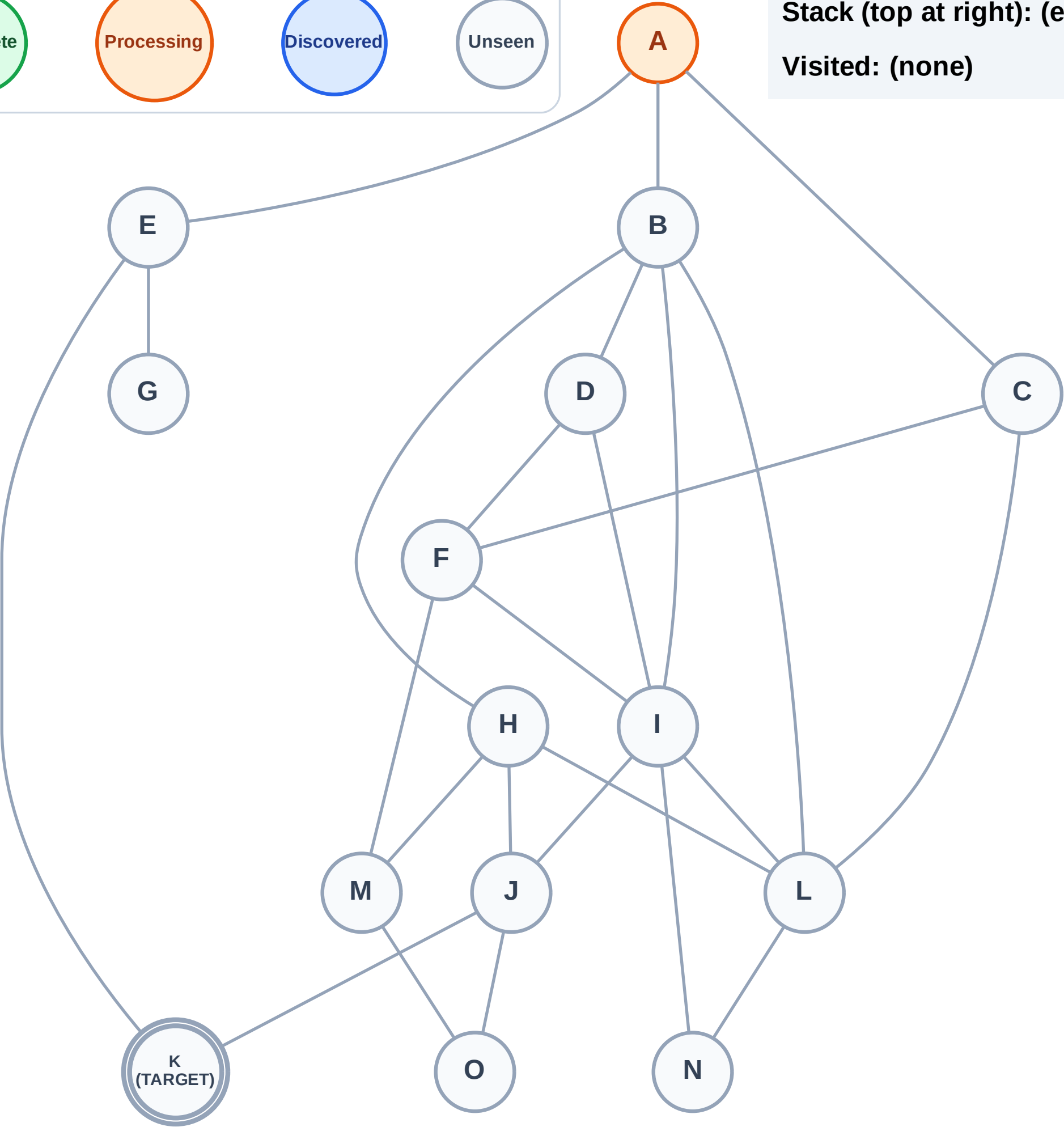
Complete

Processing

Discovered

Unseen

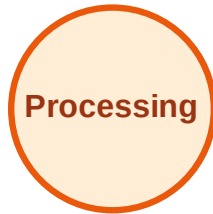
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

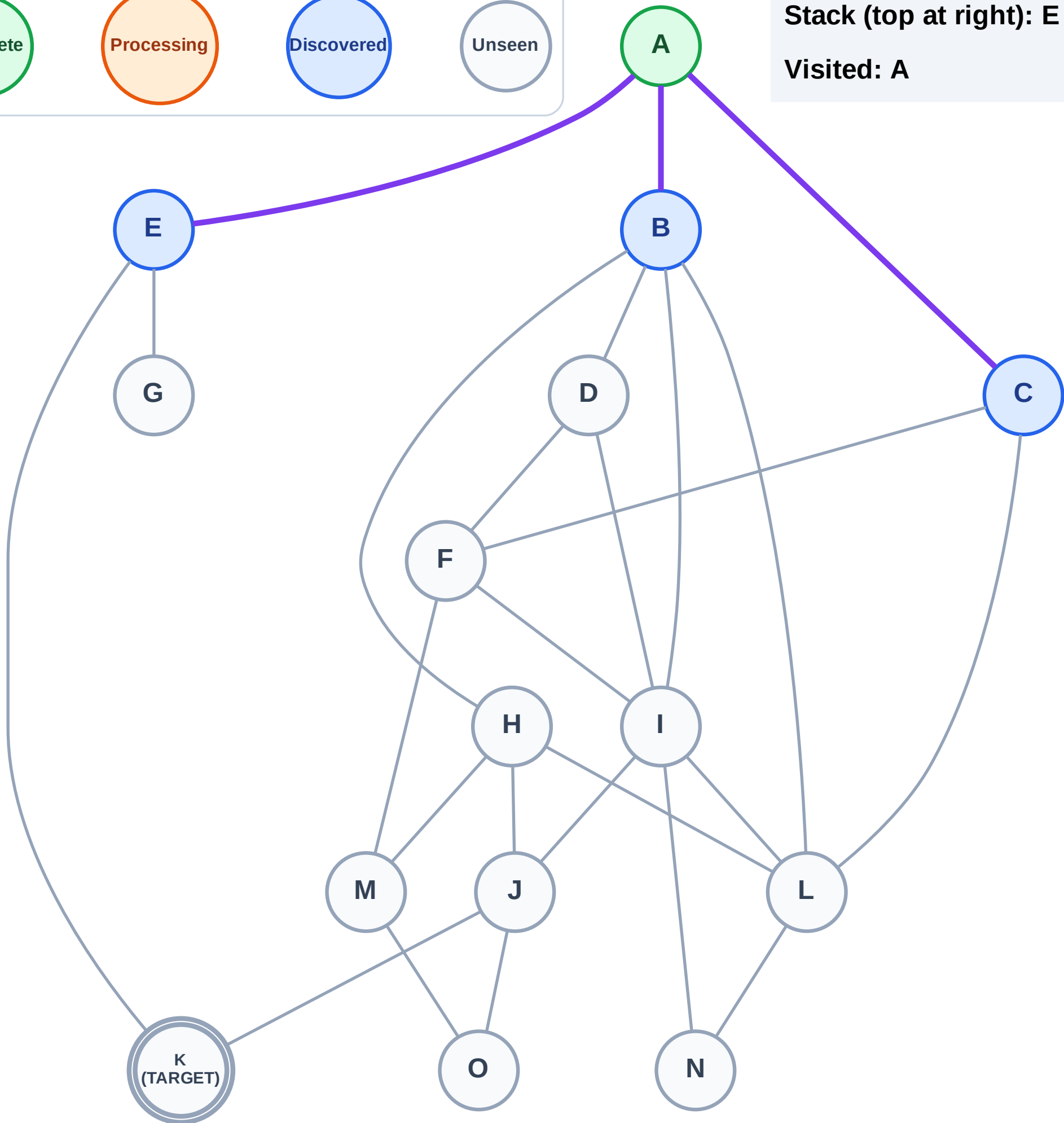
Step 3: Discover E, C, B from A

Legend



Stack (top at right): E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

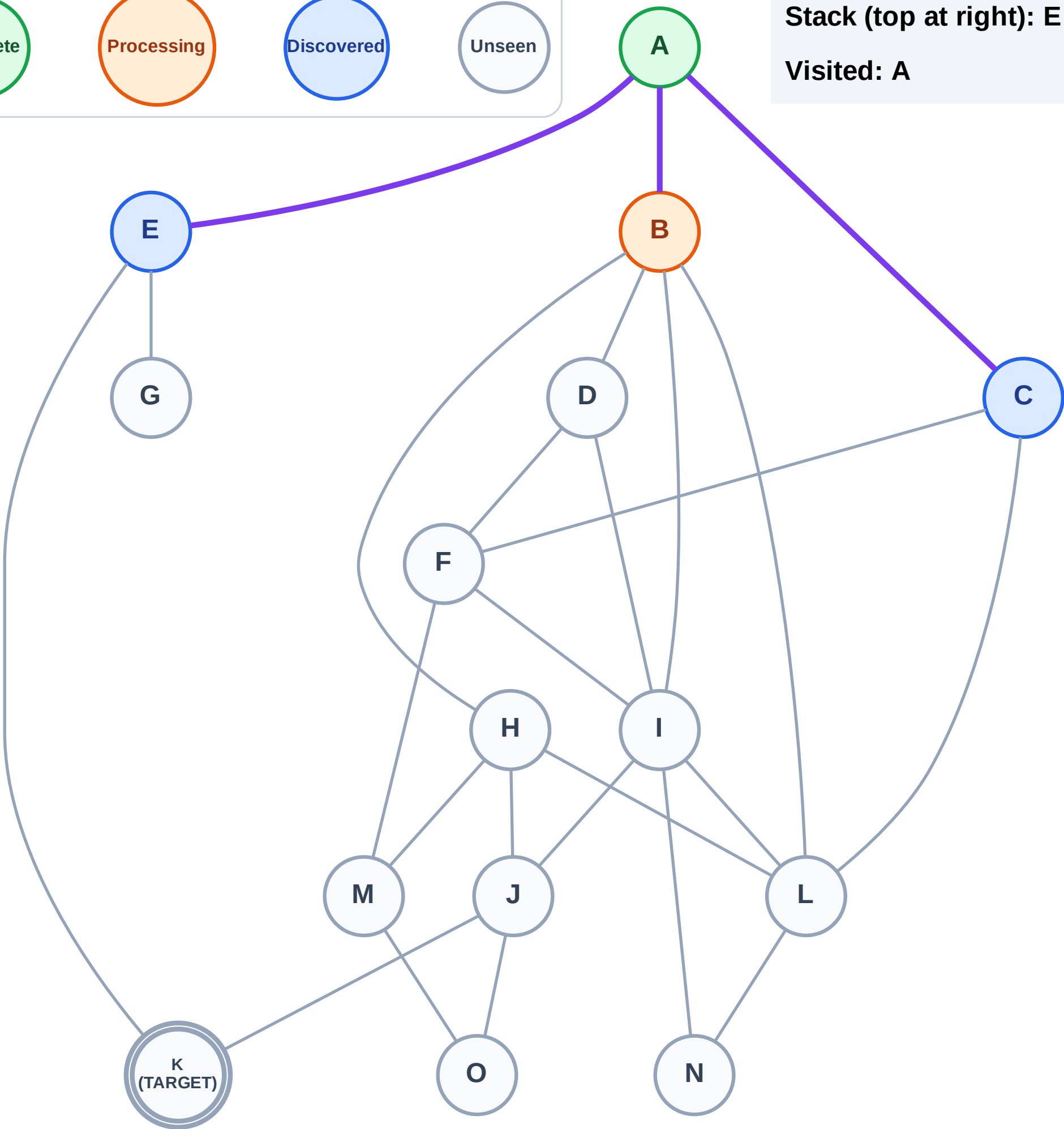
Processing

Discovered

Unseen

Stack (top at right): E | C

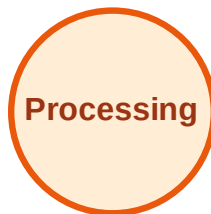
Visited: A



DFS Traversal

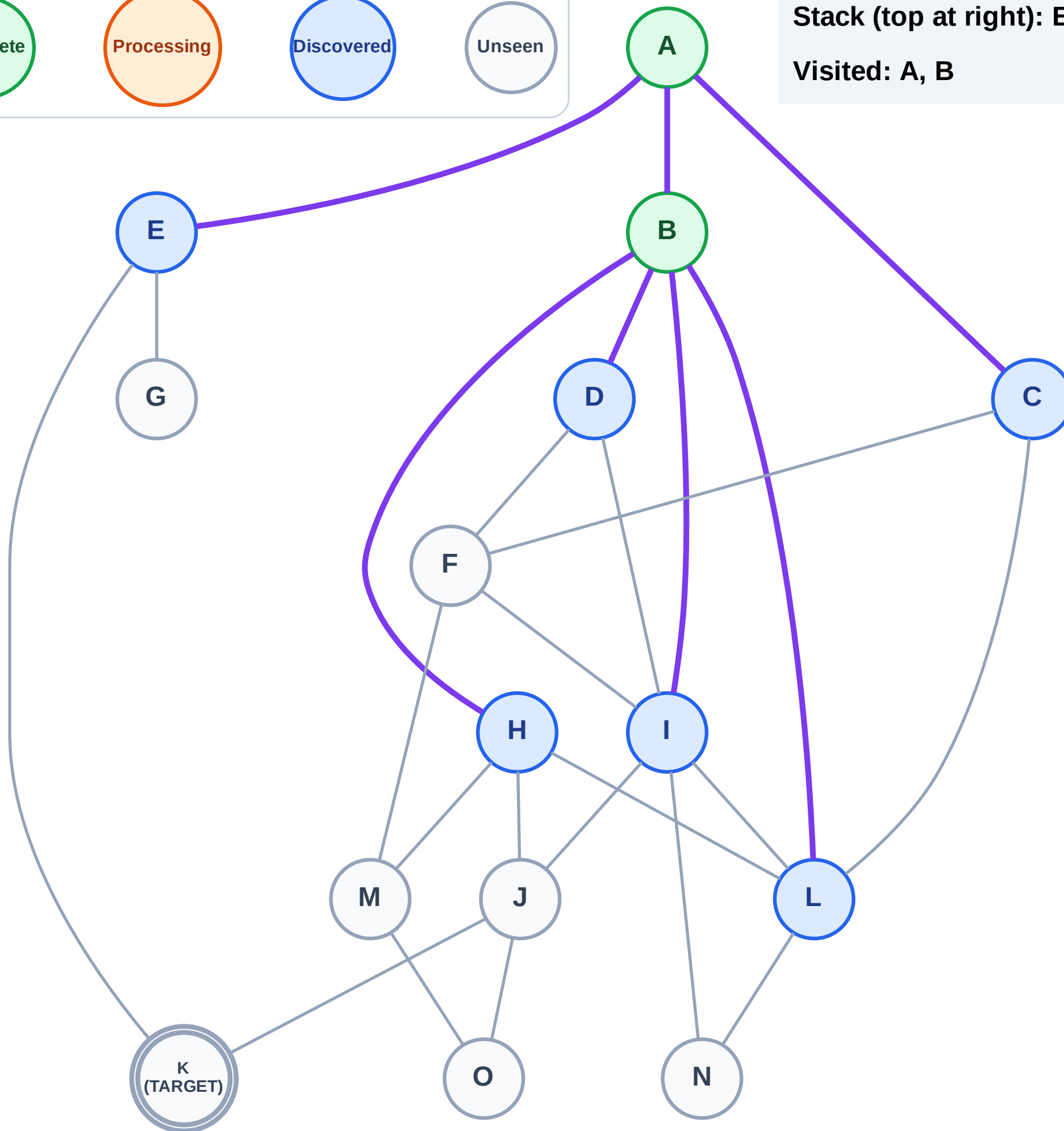
Step 5: Discover L, I, H, D from B

Legend

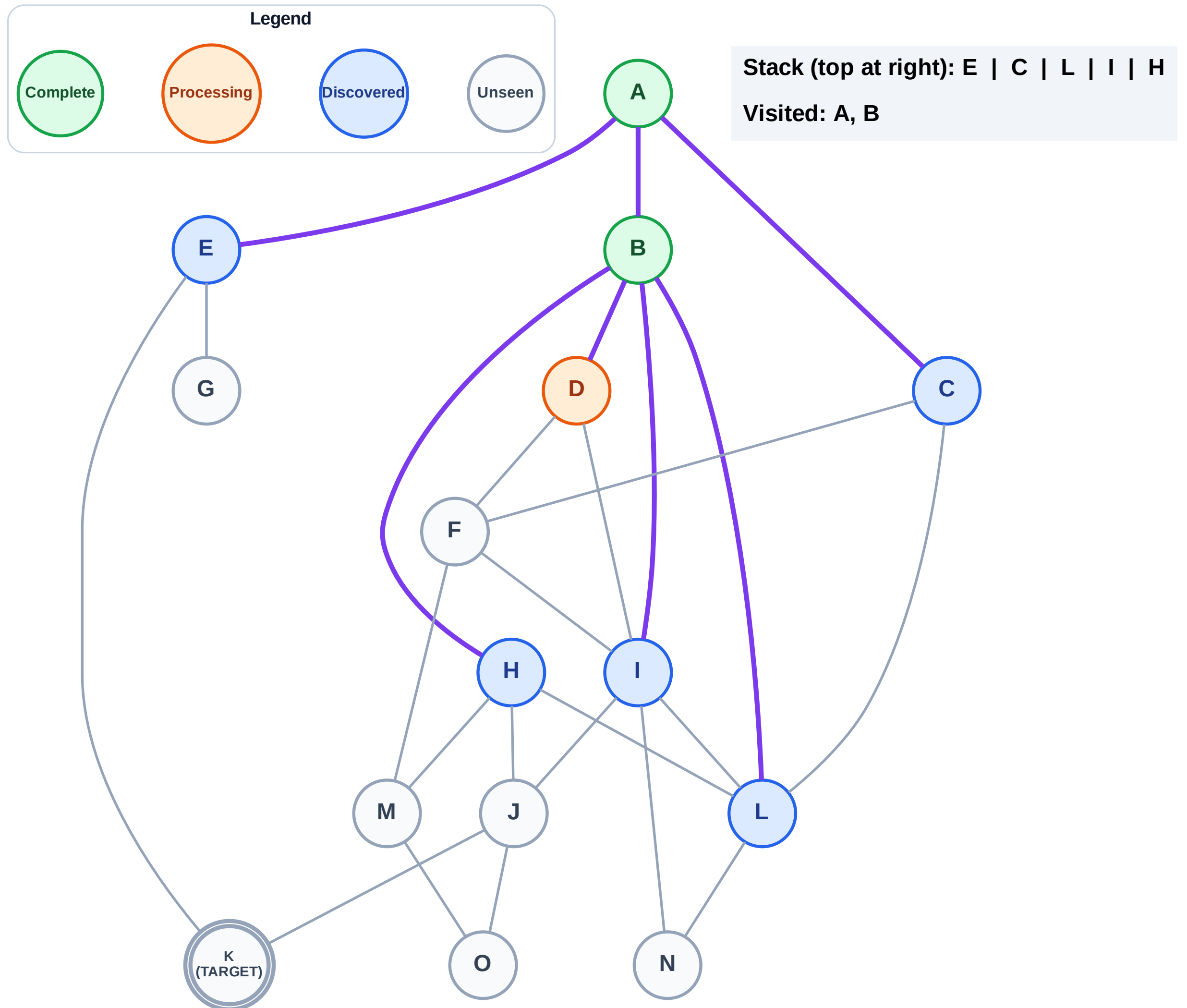


Stack (top at right): E | C | L | I | H | D

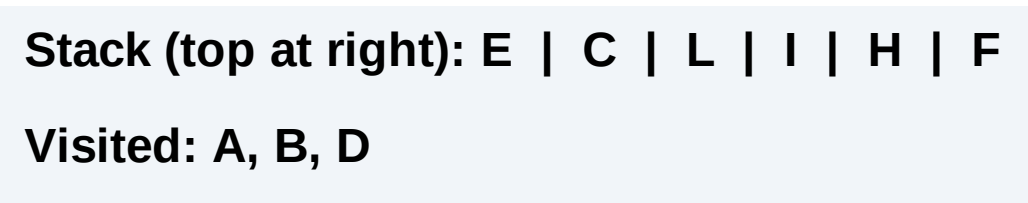
Visited: A, B



Step 6: Process node D



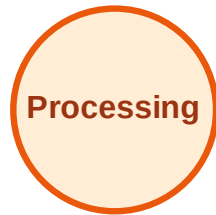
Step 7: Discover F from D



DFS Traversal

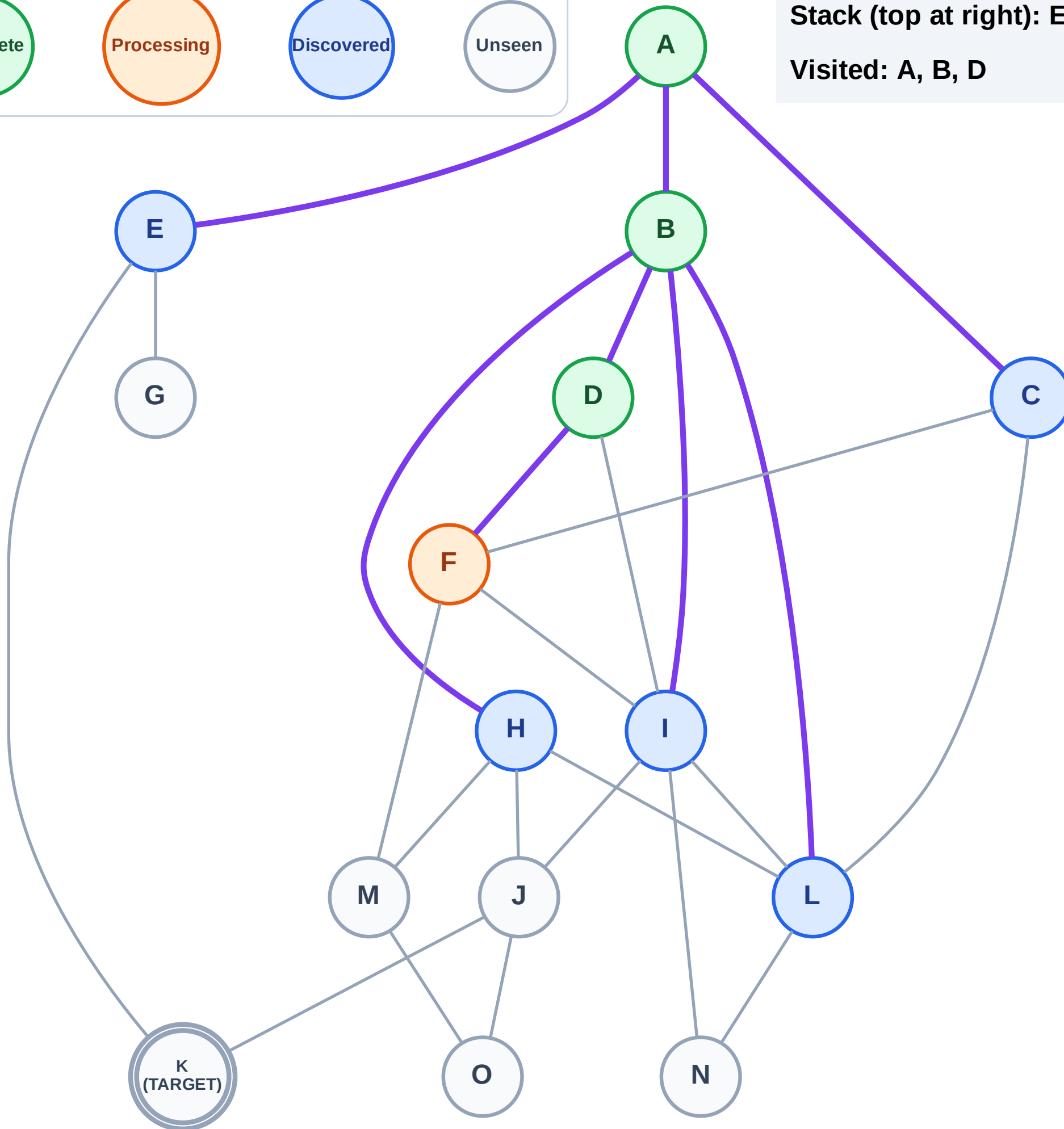
Step 8: Process node F

Legend



Stack (top at right): E | C | L | I | H

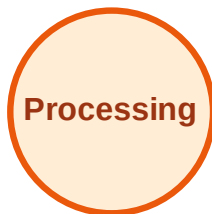
Visited: A, B, D



DFS Traversal

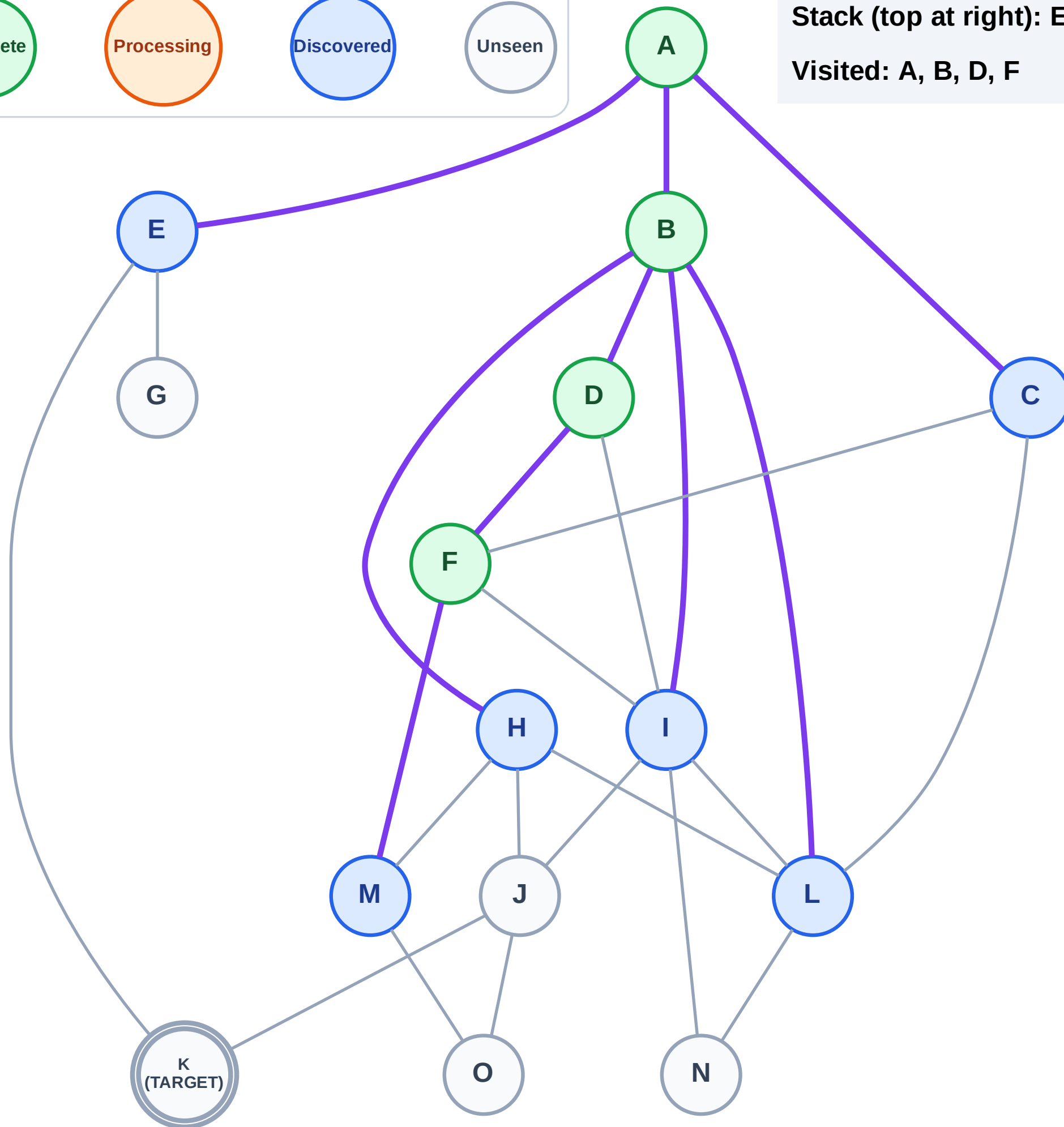
Step 9: Discover M from F

Legend



Stack (top at right): E | C | L | I | H | M

Visited: A, B, D, F



DFS Traversal

Step 10: Process node M

Legend

Complete

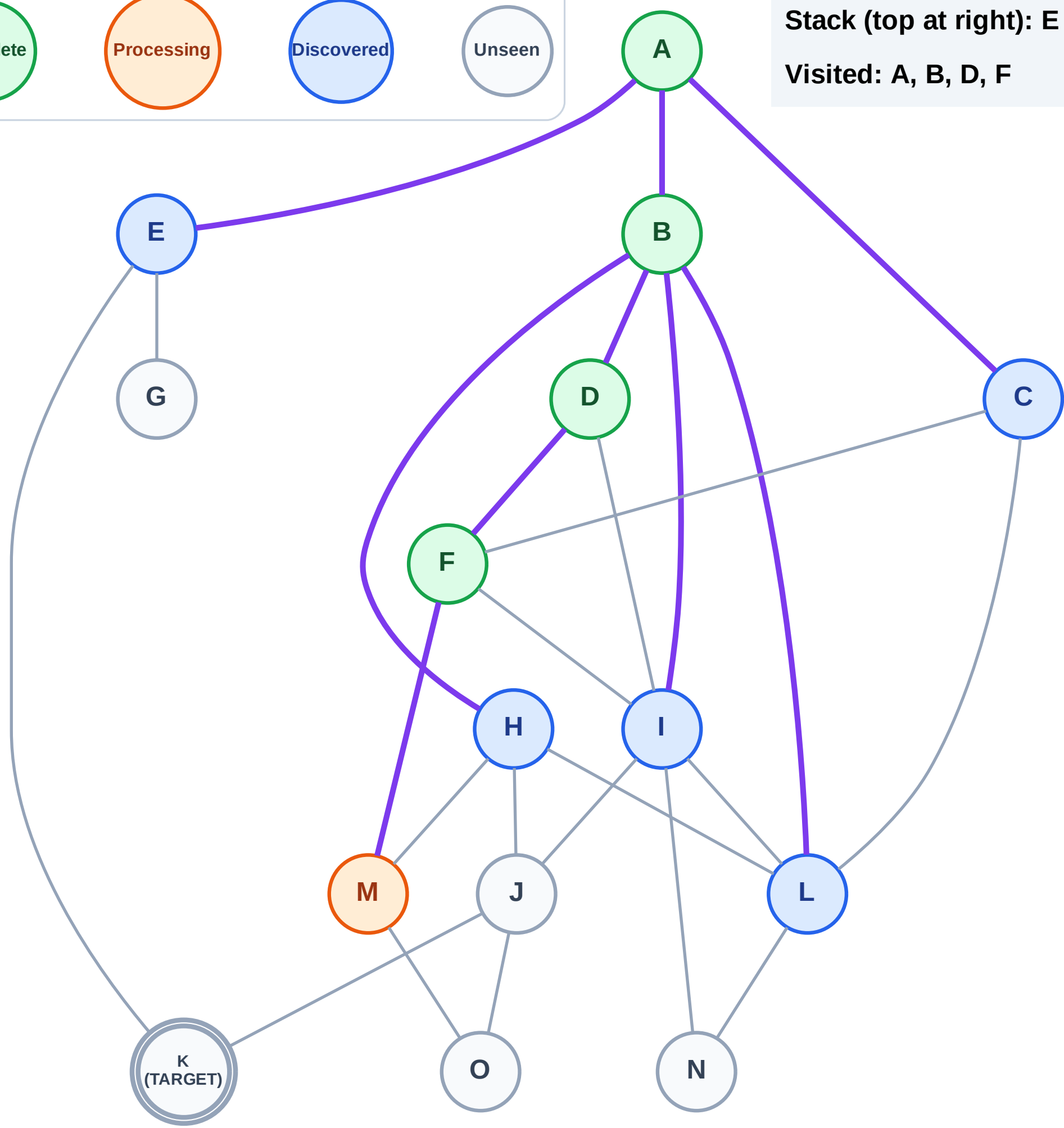
Processing

Discovered

Unseen

Stack (top at right): E | C | L | I | H

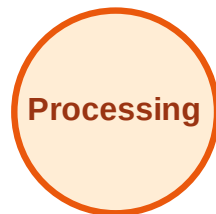
Visited: A, B, D, F



DFS Traversal

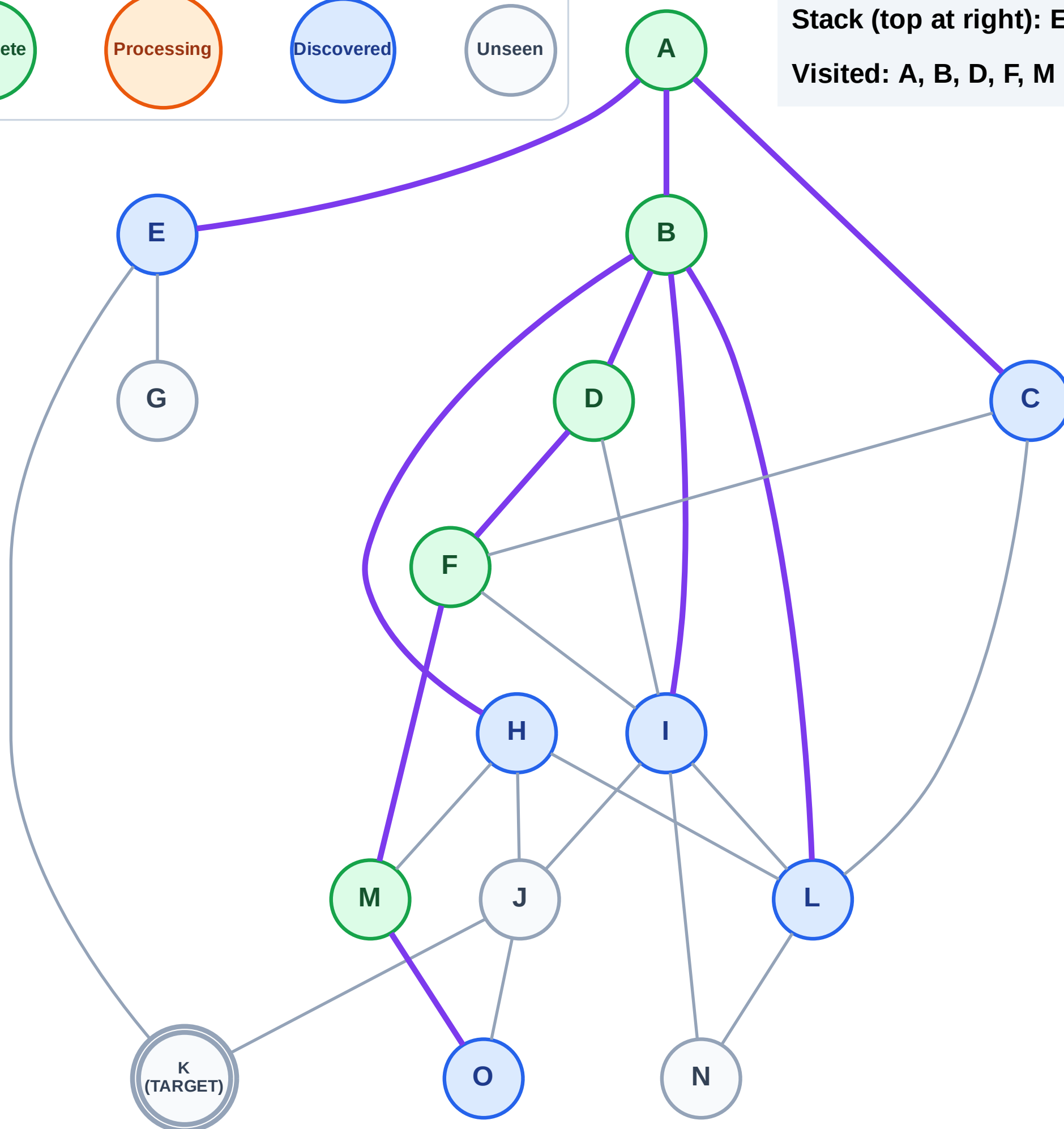
Step 11: Discover O from M

Legend



Stack (top at right): E | C | L | I | H | O

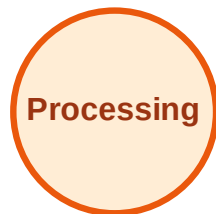
Visited: A, B, D, F, M



DFS Traversal

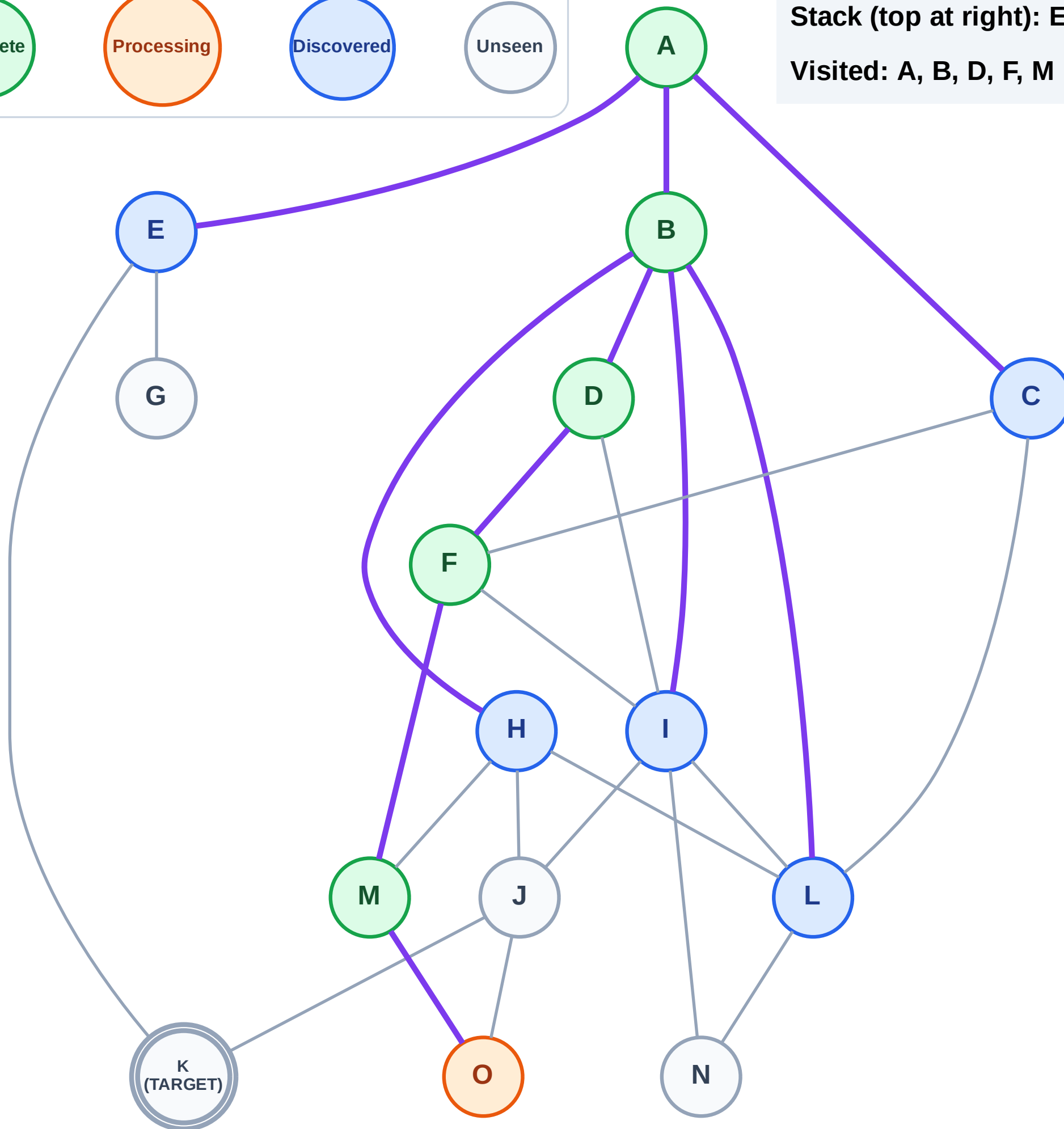
Step 12: Process node O

Legend



Stack (top at right): E | C | L | I | H

Visited: A, B, D, F, M



DFS Traversal

Step 13: Discover J from O

Legend

Complete

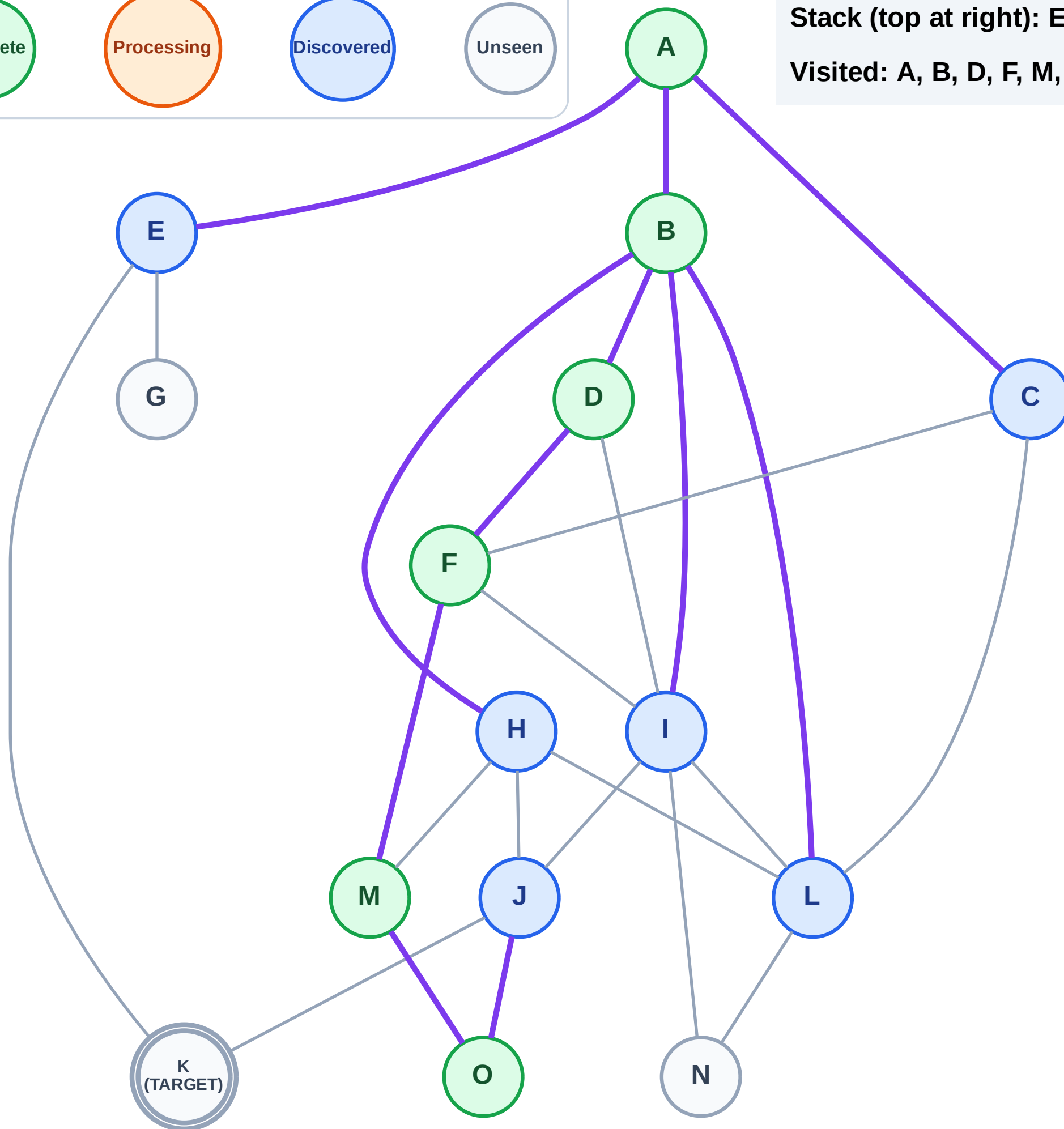
Processing

Discovered

Unseen

Stack (top at right): E | C | L | I | H | J

Visited: A, B, D, F, M, O



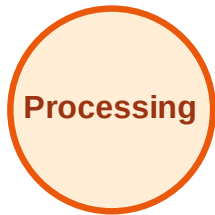
DFS Traversal

Step 14: Process node J

Legend



Complete



Processing



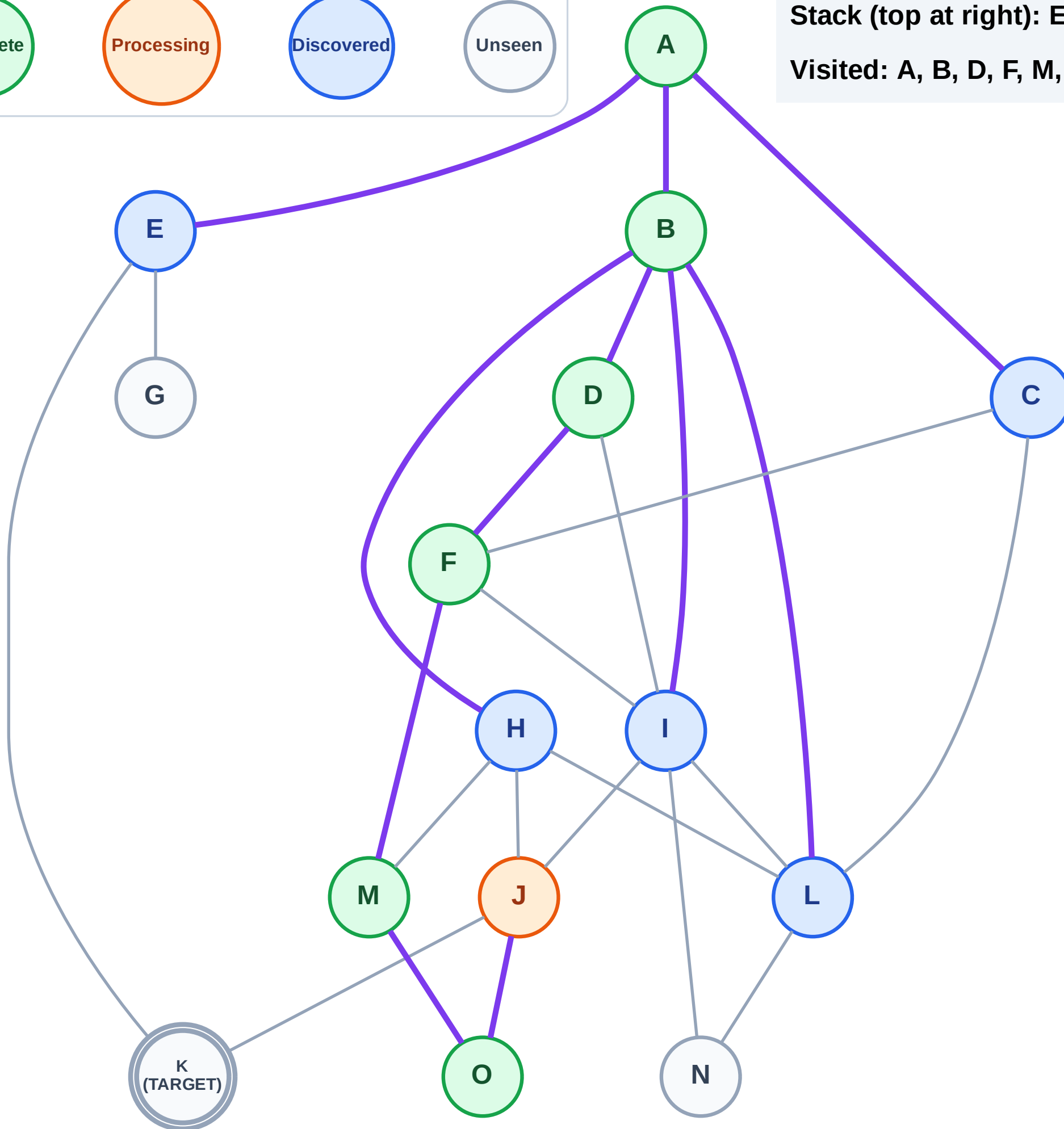
Discovered



Unseen

Stack (top at right): E | C | L | I | H

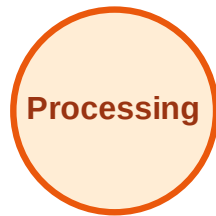
Visited: A, B, D, F, M, O



DFS Traversal

Step 15: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (7 hops): A → B → D → F → M → O → J → K

Legend



Stack (top at right): E | C | L | I | H

Visited: A, B, D, F, J, K, M, O

